

Activation Sensitivity Analysis

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Flip Rates Comparison

4-bit quantization of Dream-7B-Base and LLaDA-8B-Base. The numbers indicate the % of correct and incorrect answers of the baseline model that changed.

Model	ARC-Challenge	ARC-Easy	PIQA	Winogrande	MMLU	HellaSwag
LLaDA-8B-Base (SINQ)	3.16	2.90	1.52	5.68	8.22	0.92
LLaDA-8B-Base (AWQ)	3.75	3.24	1.58	6.08	8.80	1.18
LLaDA-8B-Base (GPTQ)	4.70	4.04	2.50	6.24	7.23	1.63
Dream-7B-Base (SINQ)	6.23	3.41	3.37	5.68	—	—
Dream-7B-Base (AWQ)	6.23	3.24	2.39	5.13	—	—
Dream-7B-Base (GPTQ)	5.20	3.33	2.61	5.84	—	—

Flip Rates Comparison

4-bit quantization of Dream-7B-Base and LLaDA-8B-Base. The first/second number indicates the % of correct/incorrect answers of the baseline model that changed.

Model	ARC-Challenge	ARC-Easy	PIQA	Winogrande	MMLU	HellaSwag
LLaDA-8B-Base (SINQ)	1.88 / 1.28	1.60 / 1.30	1.03 / 0.49	2.84 / 2.84	3.85 / 4.37	0.64 / 0.28
LLaDA-8B-Base (AWQ)	2.90 / 0.85	1.60 / 1.64	0.82 / 0.76	3.24 / 2.84	4.56 / 4.24	0.83 / 0.35
LLaDA-8B-Base (GPTQ)	2.65 / 2.05	1.94 / 2.10	1.41 / 1.09	3.08 / 3.16	4.04 / 3.19	1.20 / 0.43
Dream-7B-Base (SINQ)	3.67 / 2.56	2.06 / 1.35	1.85 / 1.52	3.39 / 2.29	—	—
Dream-7B-Base (AWQ)	4.10 / 2.13	2.06 / 1.18	1.47 / 0.92	3.08 / 2.05	—	—
Dream-7B-Base (GPTQ)	4.01 / 1.19	2.19 / 1.14	1.36 / 1.25	3.39 / 2.45	—	—

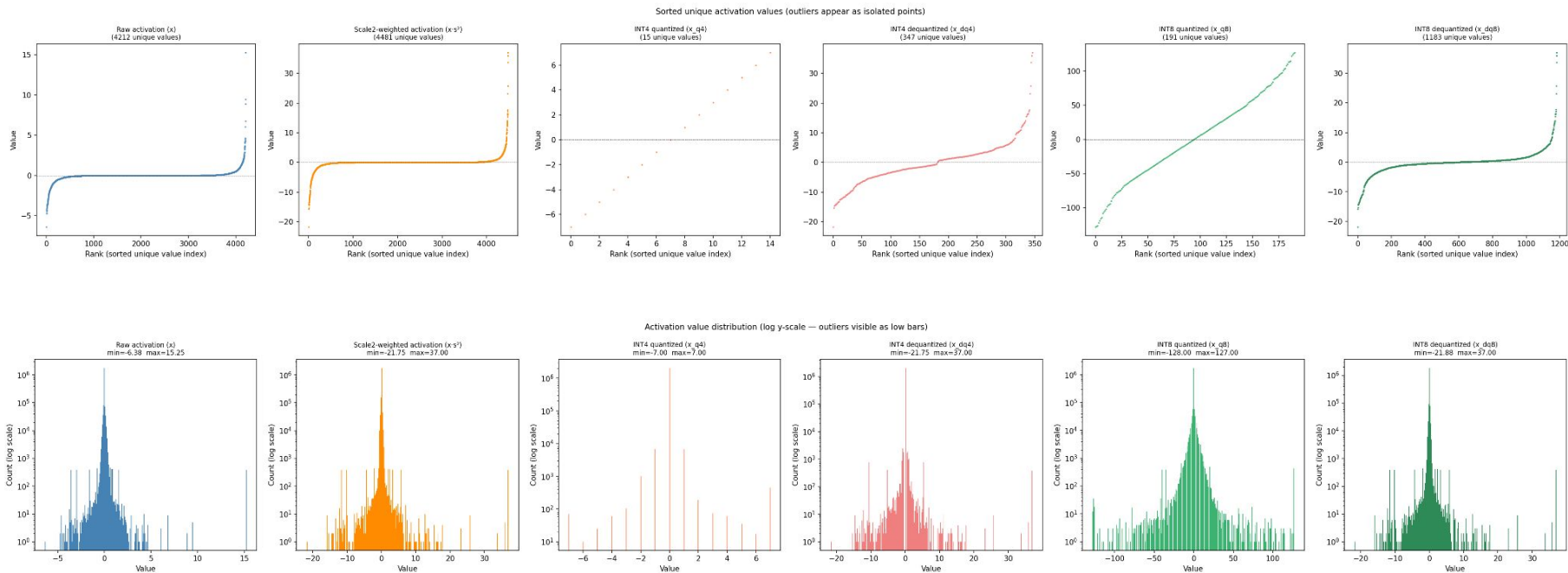
Absmax Quantization

Activations are scaled to align with SING quantization

Absmax quantization is applied to each token of these scaled activations by:

1. Finding the largest absolute value across all features for a token
2. Calculating the scale using this largest values
3. Dividing every feature of that token by the scale
4. Rounding to nearest integer

Analyzing Absmax Dynamic Quantization

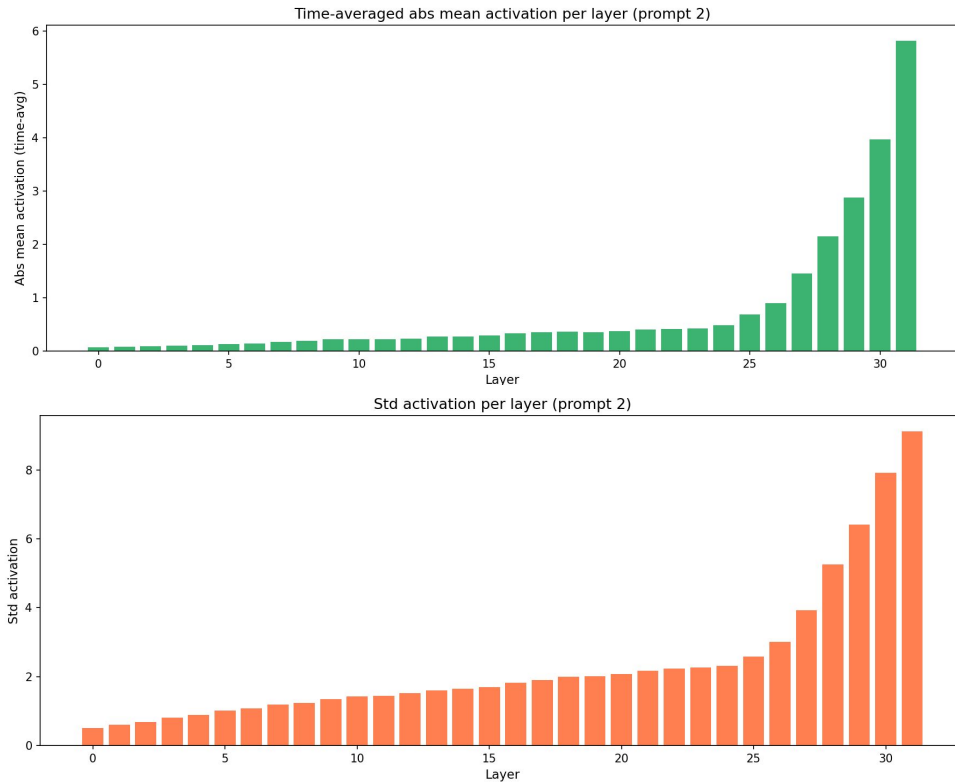


Analyzing Activation Sensitivity Across Timesteps

- The goal is to see how much the activations change between diffusion timesteps and between layers of the network
- Activation distributions are collected per layer across timesteps and the following metrics are calculated for each layer

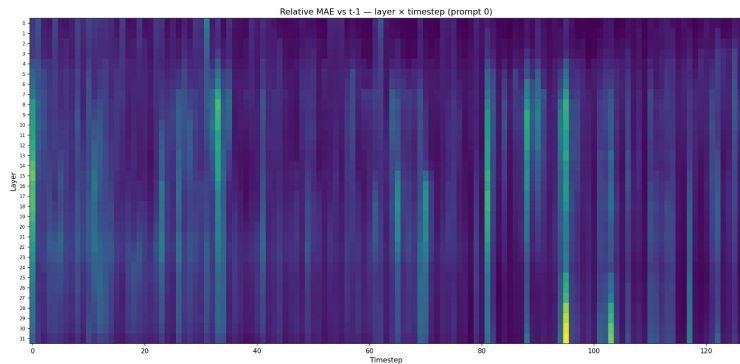
Metric	What it measures
Mean activation	Overall activation magnitude per layer
Std of activations	Dynamic range / spread of activations
Absolute mean	Mean absolute activation magnitude at each step
MAE	Average per-element change between consecutive steps
Max absolute difference	Worst-case element change between consecutive steps
Relative MAE	MAE normalized by previous step's magnitude
Relative max difference	Max change normalized by previous step's max magnitude
Cosine similarity	Directional stability of activations

Absolute Mean and Std for Activations

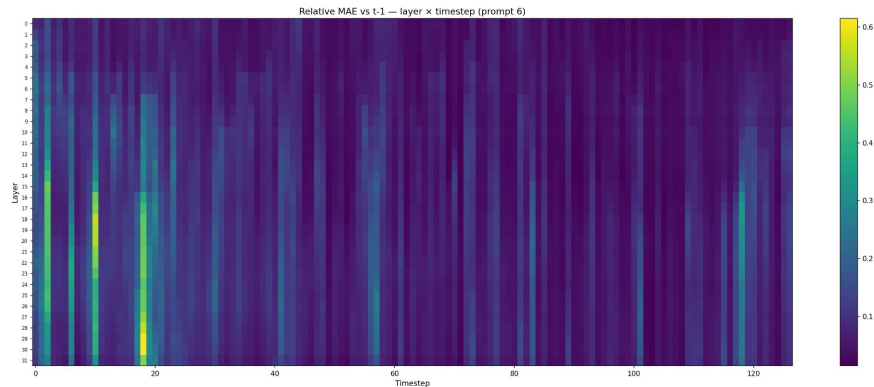


Relative Mean Absolute Error (MAE)

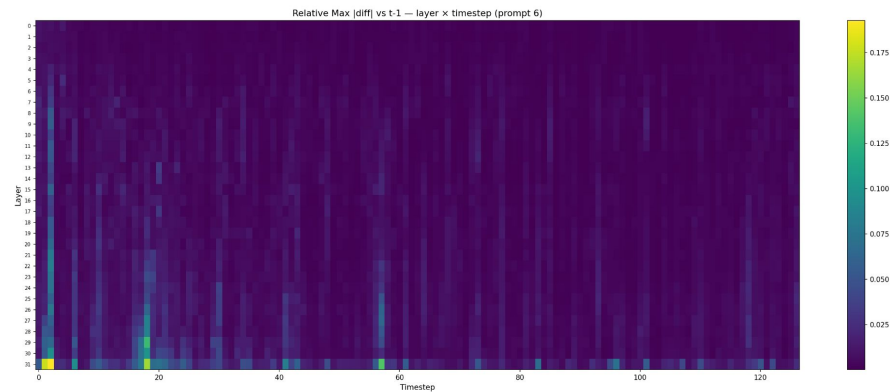
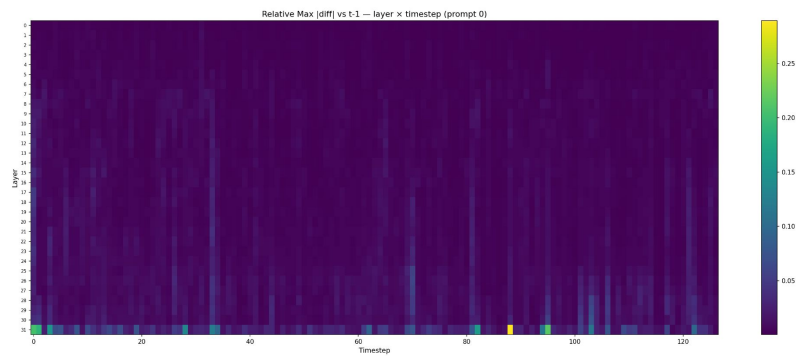
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"block_length": 32,  
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    "How many hours will it take her to read 120 pages? "  
    "Answer:"  
)
```



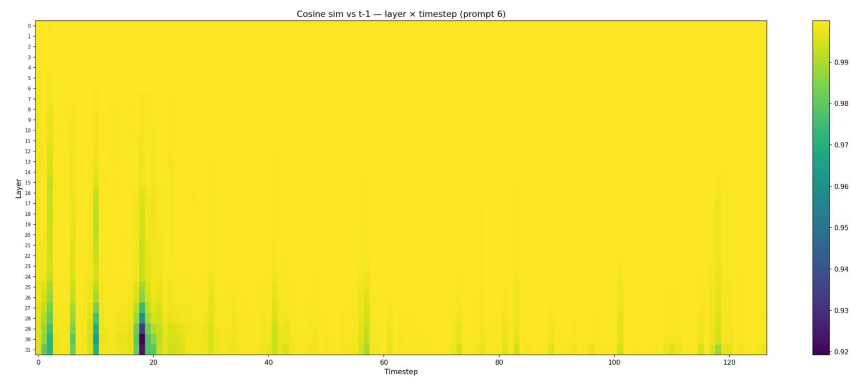
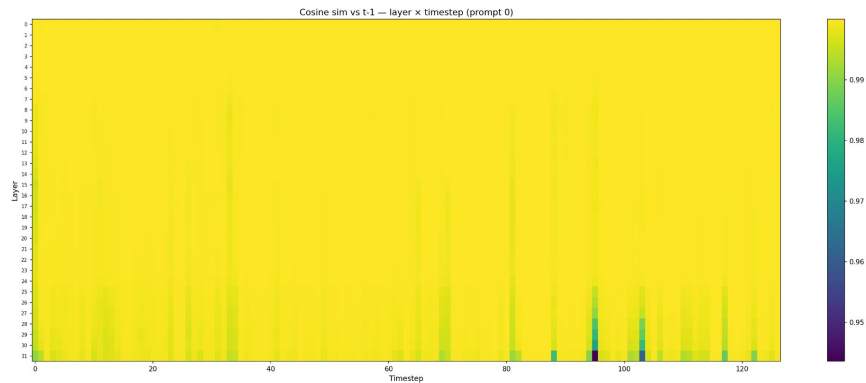
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"name": "arc_challenge",  
"block_length": 128,  
"text": (  
    "A student pushes a book across a table at constant velocity. "  
    "Explain why the book moves at constant velocity using Newton's laws of motion."  
)
```



Relative Max Difference



Cosine Similarity



Key Insights & Plans for next week

- The activation scales are increasing monotonically across the model layers
- Activations are directionally stable but their magnitudes fluctuate across layers
- Last layer (layer 31) is different from every other layer

Next week Plan

- Apply different quantization techniques to different layers, earlier layers are more stable, while later layers are more noisy